Predicting Income Level Using Logistic Regression and Random Forest

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INTRODUCTION

With the rapid development of data mining and machine learning technologies, classification models have been widely applied in socioeconomic analysis. The Adult dataset, released by the U.S. Census Bureau, is a well-known benchmark dataset used to predict whether income exceeds $50K/yr based on census data. Analyzing this dataset not only demonstrates model performance but also provides insight into the social factors that influence income levels.

This study compares two representative classifiers, Logistic Regression and Random Forest, to evaluate their effectiveness in income classification tasks. Logistic Regression is a linear model with strong interpretability, while Random Forest is a nonlinear ensemble model capable of capturing complex interactions among features. By comparing their predictive performance, computational efficiency, and generalization ability, this report aims to identify which model performs better for the Adult dataset.

METHODS OR PROCEDURES

1 Data Preparation and Preprocessing
The Adult dataset contains 45,175 samples and 14 features, including both numerical and categorical variables. The data preprocessing steps were as follows:

1.1 Handling Missing Values
Missing values were originally represented by “?”. These were replaced with “NaN” for proper recognition by Pandas. As the missing rate was relatively low (mainly in work-class, occupation, and native-country), rows containing missing values were removed. Duplicate entries were also deleted to ensure data integrity.

1.2 Label and Feature Cleaning
First, removed redundant characters in the target variable (e.g., ‘>50K.’change to ‘>50K’). Second, converted the target variable income into binary form (‘>50K’ change to ‘1’, ‘<=50K’ change to ‘0’). Third, dropped the variables ‘education’ and ‘fnlwgt’ which aren’t useful for classification.

1.3 Encoding Categorical Variables
Applied one-hot encoding to discrete features (workclass, marital-status, occupation, relationship, race, sex). Convert these into numerical formats easy for models to handle.

1.4 Clustering countries
To reduce the dimensionality of the categorical feature ‘native-country’, countries were grouped based on their socio-economic characteristics. K-Means clustering was applied to average national indicators, and the optimal number of clusters was determined using the Elbow Method.

1.5 Handling Class Imbalance
Since high-income individuals account for less than one-fourth of the total samples, the Synthetic Minority Oversampling Technique (SMOTE) was applied to address the issue of class imbalance and to improve model performance on the minority class.

2 Model Construction and Configuration
2.1 Feature Scaling
Logistic Regression optimizes its parameters by minimizing a loss function through gradient descent. Since this optimization process depends on the magnitude of the gradients, the scale of input features can significantly affect convergence. Large differences in feature ranges may lead to slow or unstable convergence. Therefore, standardization was applied to numerical variables (age, education-num, hours-per-week, capital-gain, and capital-loss) to accelerate convergence and improve model stability.

In contrast, Random Forest constructs decision trees by identifying optimal split points. Because the determination of split points is not affected by the absolute scale of the input features, Random Forest is insensitive to feature scaling. Thus, the model was trained using the original numerical data.

2.2 Logistic Regression
This linear model determines the relationship between features and the target through learned coefficients.

2.3 Random Forest
An ensemble of decision trees using default parameters. The model captures nonlinear relationships and feature interactions through bagging and random feature selection.

3 Model Training and Evaluation Metrics
The dataset was split into 70% training data and 30% testing data.
Performance was evaluated using the following metrics: Accuracy, Precision, Recall, F1-score, ROC curve and AUC (Area Under the Curve), Training and prediction time.

RESULTS

1 Model Performance
Table 1. Classification Performance Metrics for Random Forest and Logistic Regression Models

Model	Accuracy	Precision	Recall	F1-Score
Random Forest	0.884	0.874	0.900	0.886
Logistic Regression	0.835	0.854	0.866	0.855

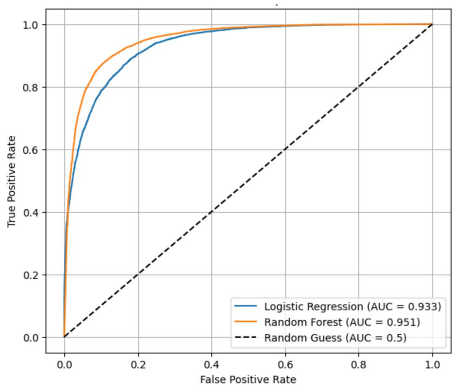
According to Table 1, Random Forest model outperformed Logistic Regression in all major metrics, indicating superior ability to identify high-income individuals.

2 Training and Prediction Time
Table 2. Training and Prediction Time Comparison of Random Forest and Logistic Regression Models

Model	Training Time (s)	Prediction Time (s)
Random Forest	6.743	0.521
Logistic Regression	1.442	0.012

According to Table 2, Logistic Regression exhibited significantly higher computational efficiency. Its training speed was about 4 times faster than Random Forest, and its prediction speed was over 40 times faster.

3 ROC Curve and AUC Analysis
Figure 1. ROC Curve Comparison between Logistic Regression and Random Forest



Both models demonstrated strong discriminative ability, as their ROC curves were well above the random baseline (AUC = 0.5).Logistic Regression's AUC ≈ 0.933 while Random Forest's AUC ≈ 0.951.

The ROC curve of Random Forest enveloped that of Logistic Regression, showing consistently higher true positive rates at equivalent false positive rates.
This indicates that Random Forest offers more stable classification performance across different decision thresholds.

DISCUSSION

The experimental results reveal that Random Forest achieves higher accuracy, F1-score, and AUC compared to Logistic Regression, confirming its advantage in modeling complex and nonlinear relationships among socioeconomic factors. Logistic Regression, although slightly less accurate, remains a valuable model for its interpretability, stability, and computational efficiency.

Random Forest effectively captures feature interactions such as between education level and working hours, which are often nonlinear. This enhances its ability to distinguish between income groups. However, it comes at the cost of higher training and inference time, as well as reduced interpretability.

Both models demonstrate the critical importance of proper data preprocessing. Standardization, categorical encoding, and removal of redundant features significantly improved model robustness and training convergence.

CONCLUSION

In conclusion, this study compared the performance of Logistic Regression and Random Forest on the Adult income dataset.

During the training, Random Forest achieved better overall performance, with higher accuracy, recall, and AUC while Logistic Regression offered faster training and prediction times and stronger interpretability.

Therefore, If prediction performance is the primary goal, Random Forest is the preferred choice. If model transparency and computational efficiency are prioritized, Logistic Regression remains an effective solution.

REFERENCES

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